5. Methodology

5a. Methodology - Program 1: Electricity Bill Calculator

1. **Program Design**
   - Developed a menu-driven interface with reusable box-drawing functions and an ASCII-art banner.
2. **Billing Logic Implementation**
   - Implemented calc() to calculate slab-wise electricity charges, apply discounts, and add a fixed service charge.
3. **Data Handling & Validation**
   - Implemented Stored user records in an in-memory dictionary and validated unit input using exception handling.

5b. Methodology - Program 2: ATM Transaction Simulator

1. **Program Structure**
   - Created a boxed menu interface using reusable helper functions and an ASCII-art banner.
2. **Account Data Modelling**
   - Stored account details in a Python dictionary containing card number, PIN, and balance.
3. **Core Transaction Logic**
   - Implemented authentication, balance enquiry, withdrawal, deposit, and PIN change with necessary validation.
4. **Menu-Driven Control Flow**
   - Built a continuous menu system and tested various valid and invalid transaction scenarios.

6. Expected Outcomes / Deliverables

6a. Program 1: Electricity Bill Calculator

- A functional slab-wise electricity bill calculator.
- Accurate discount and fixed-charge calculation.
- Menu-driven interface with user record management.
- Scope for future enhancements.

6b. Program 2: ATM Transaction Simulator

- A functional ATM simulator with authentication and transactions.
- Input validation for users and transactions.
- User-friendly boxed terminal interface.
- Scope for future enhancements such as persistent storage and account security.